

FPGA / RTL / DV / PHYSICAL SIGNOFF

AES-GCM

RTL Engineering Report

從SystemVerilog架構、握手與安全控制，一路追溯到175 MHz routed DCP與FPGA板級NIST驗證。

Artix-7 XC7A100T

Vivado 2021.1

175 MHz

RTL BASELINE

4246ebc7...578c11

PRODUCTION TOP

arty_a7_100t_aes_gcm_uart_rs
p_top

SCOPE

RTL → Simulation → Implementation → Board

CURRENT FULL-BOARD STATUS

IMPLEMENTED · ROUTED · VERIFIED

主管先看什麼：RTL不是檔案清單，而是三層證據

01

Structure

模組邊界、資料流、clock/reset與可替換介面。

- Board wrapper與crypto core分離
- AXI-style ready/valid contract
- AES、GHASH、Key Context、FIFO分工

02

Mechanism

RTL如何達成吞吐率、時序與安全不變條件。

- 三個AES service instances
- 16-bit digit-serial GHASH
- authenticated plaintext release

03

Evidence

每個主張都連到simulation、DCP、RPT或board log。

- WNS +0.029 ns · TNS 0
- NIST 47,250 / 47,250
- DRC與route errors 0

ENCRYPT CORE THROUGHPUT

1.065

Gbit/s · simulation

DECRYPT CORE THROUGHPUT

1.052

Gbit/s · simulation

FINAL ROUTED CLOCK

175

MHz · 5.714 ns

HARDWARE NIST

0

failures / 47,250

Production RTL地圖：14個檔案，6個責任域

engrenring/rtl/source/

board/ board top · UART/RSP bridge · uart_rx/tx

stream_axi/ AXI-Lite registers · output skid

stream_core/ transaction controller · stream FIFO

stream_aes/ AES engines · key context

stream_ghash/ production ghash16

aes_core/ S-box primitive

主報告只追三條線

Protocol boundary

UART/RSP → AXI-Lite / AXI-Stream

Crypto datapath

AES-CTR ↔ GHASH ↔ tag compare

Security lifecycle

accept → authenticate → release / abort / zeroize

`tb\_ghash16.sv`是testbench，不列入production RTL。

RTL最後修改COMMIT

4246ebc7a72ce8eb1a49d56969aae6e151578c11

分層架構：板級協定不污染crypto core



Interface contract : registered READY切斷wide feedback cone

```
1 // aes_gcm_axi_top.v:156-201
2 assign s_axis_tready = s_axis_tready_r && !zeroize_pulse;
3 wire public_input_fire = s_axis_tvalid && s_axis_tready;
4
5 wire [2:0] inputq_count_next = {1'b0, inputq_count} +
6                               (public_input_fire ? 3'd1 : 3'd0)
7                               (inputq_pop ? 3'd1 : 3'd0);
8
9 always @(posedge aclk) begin
10     if(!aresetn || zeroize_pulse) begin
11         input_frame_credits ≤ 2'd0;
12         s_axis_tready_r ≤ 1'b0;
13     end else begin
14         input_frame_credits ≤ input_frame_credits_next;
15         if(core_zeroize_busy)
16             s_axis_tready_r ≤ 1'b0;
17         else
18             s_axis_tready_r ≤ (input_frame_credits_next ≠ 0) &&
19                               (inputq_count_next < 2);
20     end
21 end
```

01 Same-edge admission

`valid && ready`是唯一accept事件。

02 Predictive occupancy

用next count決定下一cycle READY，避免overflow。

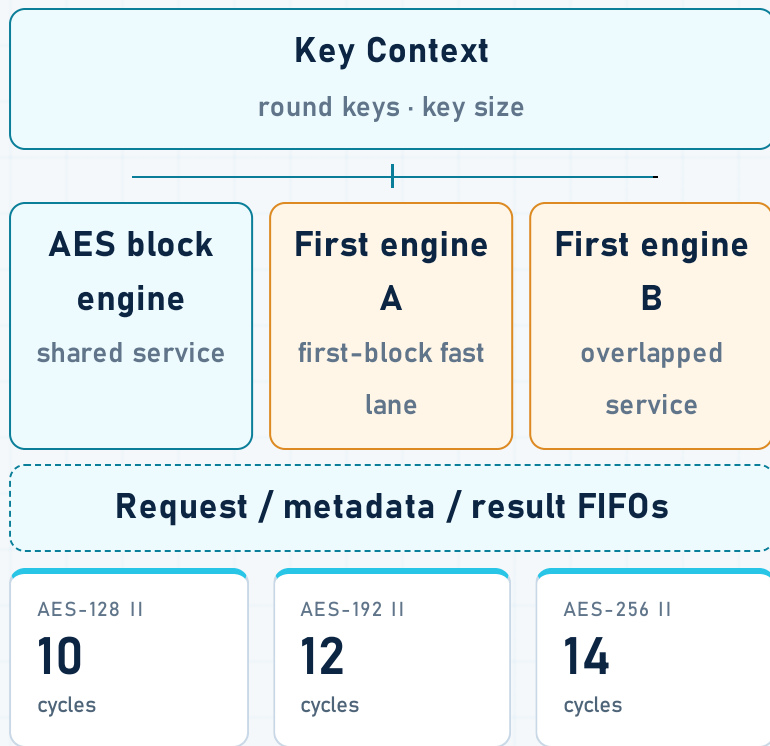
03 Timing locality

READY註冊化，不把core-wide狀態組合回public port。

04 ZEROIZE priority

command pulse同cycle直接gate READY。

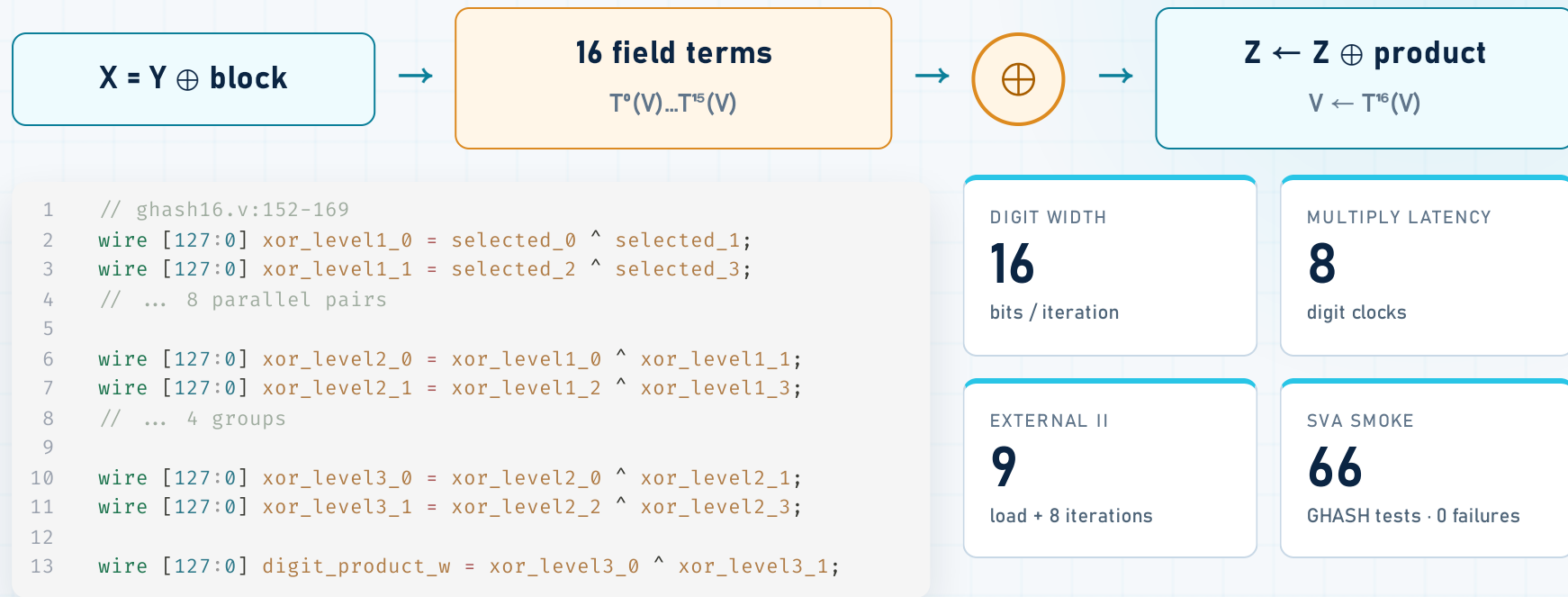
AES service : 一組共享round-key context , 三個engine instances



```
1 // aes_gcm_stream_core.v:314-335
2 aes_block_engine u_aes_engine (
3     .clk(clk), .rst_n(rst_n && !zeroize),
4     .key_size(expanded_key_size),
5     .round_keys(round_keys),
6     .block_in(aes_engine_in),
7     .block_out(aes_engine_out)
8 );
9
10 aes_first_block_engine u_aes_first_engine (
11     .clk(clk), .rst_n(rst_n && !zeroize),
12     .round_keys(round_keys),
13     .block_in(aes_prod_data[267:140]),
14     .block_out(aes_first_out)
15 );
16
17 aes_first_block_engine u_aes_second_engine (
18     .clk(clk), .rst_n(rst_n && !zeroize),
19     .round_keys(round_keys),
20     .block_in(aes_prod_data[267:140]),
21     .block_out(aes_second_out)
22 );
```

設計重點：複製的是服務能力，不是key schedule；round keys由單一context供應。

GHASH16 : 16-bit digit-serial , 組合邏輯刻意做成balanced XOR



Decrypt fail-closed : 先驗證tag , 再授權plaintext release

```
1 // aes_gcm_stream_core.v:2536-2576
2 if(descriptor_crypto_active_w && final_tag_ready &&
3    all_cipher_blocks_ready && !result_valid_r) begin
4     if(rec_decrypt && tag_compare_ready_r &&
5        !plain_write_pending_valid_r) begin
6         result_data_r ≤ tag_mismatch_w ?
7             RESULT_DEC_AUTH_FAIL :
8             RESULT_DEC_AUTH_OK;
9         result_valid_r ≤ 1'b1;
10        state ≤ ST_DEC_RESULT;
11    end
12 end
13
14 if((state = ST_DEC_RESULT) && result_valid_r &&
15    m_axis_result_tready) begin
16     result_valid_r ≤ 1'b0;
17     if(result_data_r = RESULT_DEC_AUTH_OK) begin
18         plain_release_valid ≤ 1'b1;
19         plain_release_data ≤ {rec_bank, rec_data_bits};
20     end
21 end
```

Ciphertext → private plaintext banks



完整tag operands + fixed XOR/OR
compare

AUTH_OK
release token

AUTH_FAIL
no plaintext release

Board reconciliation : unauthorized plaintext = 0

安全不變條件 : plaintext的存在不等於對外可見 ; 只有AUTH_OK result被接受後才產生release permission 。

ZEROIZE與FIFO：同edge阻擋新交易，再同步清除有效狀態

BOUNDARY GATE

```
assign s_axis_tready =  
    s_axis_tready_r &&  
    !zeroize_pulse;
```

raw command pulse同cycle阻止新的public accept。

QUEUE INVALIDATION

```
stream_fifo u_input_queue (  
    .clear(zeroize_pulse),  
    // ready / valid queue  
);
```

count、pointer與registered head同步歸零，stale head不能fire。

SCRUB LIFECYCLE

```
if(first_zeroize_event_w)  
begin  
    zeroize_busy_r ≤ 1'b1;  
    zeroize_index ≤ 5'd0;  
end
```

重複ZEROIZE保持idempotent，scrub完成前不重新開放context。

T0

raw ZEROIZE

same edge

READY/VALID gated

T1

FIFO/control invalidated

scrub

16 addresses

done

abort result / clean recovery

Production sources : [aes_gcm_axi_top.v](#) · [aes_gcm_stream_core.v](#) · [stream_fifo.v](#)

Timing-aware RTL : 局部ready與registered head降低control cone

```
1 // stream_fifo.v:27-37
2 wire push = in_valid && in_ready;
3 wire pop  = out_valid && out_ready;
4
5 // Full只看count MSB, localize RAM write enable
6 assign in_ready  = !count_r[ADDR_W];
7 assign out_valid = (count_r ≠ 0);
8
9 // Head註冊化, pointer select不直達consumer mux
10 assign out_data = out_data_q;
```

Post-synthesis estimate

-0.298 ns

292 setup endpoints

+0.327 ns

Final post-route

+0.029 ns

TNS 0 · FEP 0

Final worst path

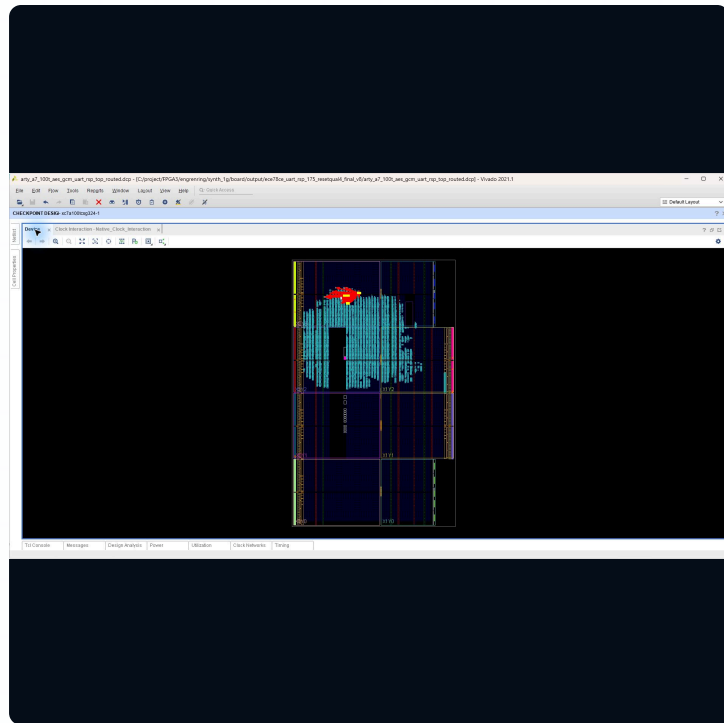
1.214 ns logic

5.544 ns data path

4.330 ns net · 78.1%

來源：stream_core/stream_fifo.v。Synthesis成功但初估setup未closure；placement、physical optimization與routing後，175 MHz final timing通過。

Code → hardware : 最終routed DCP是release基準



Vivado Device View · final routed DCP

SETUP

+0.029

WNS ns · TNS 0

HOLD

+0.044

WHS ns · THS 0

ROUTING

17,758

/17,758 · errors 0

DRC

0

violations

12,411

LUT - 19.58%

9,029

FF · 7.12%

10

RAMB18

0

DSP

Routed DCP SHA-256

A91758ED...9C016B

Power 0.461 W是Vivado vectorless estimate (Medium) , 不是實板量測。

Verification pyramid : 不同方法覆蓋不同失效模式

FPGA BOARD

47,250 / 47,250 NIST · 6 / 6 smoke

XSim NIST RSP

47,250 / 47,250 · 17,740,527 cycles

UVM 1.2

3 seeds · 24 checked · 0 failed · maintained bins 100%

SVA

37 properties · AXI/GHASH smoke · 0 assertion failures

STATIC / FORMAL SUBSET

DRC · methodology · CDC gates · 19 combinational assertions

What is closed

Full-board timing · routing · DRC · hardware NIST與security reconciliation ·

What is not claimed

Whole-design sequential formal · code/assertion coverage · measured board power · OOC PartPin ·

Release narrative：程式碼主張與證據一一對應

RTL CLAIM	MECHANISM	EVIDENCE	STATUS
無overflow的stream accept	registered READY + credits	UVM / SVA / directed	PASS
高吞吐AES-GCM	3 AES services + FIFO decoupling	1.052-1.065 Gbit/s simulation	PASS
Decrypt fail-closed	AUTH_OK-gated plaintext release	11,908 expected auth fails · leakage 0	PASS
ZEROIZE無stale output	same-edge gate + clear + scrub	SVA / directed / board reconciliation	PASS
175 MHz implementation	timing-aware RTL + physical optimization	WNS +0.029 · TNS 0	PASS
Standalone OOC reuse	approved PartPin map	identity-matched map unavailable	OPEN

主管結論

RTL架構、security control、implementation與板級功能形成可追溯閉環；目前release限制被明確隔離，沒有用一種測試替代另一種signoff。

Appendix

原始RTL、Vivado native evidence與verification summaries

Appendix A · Source與evidence索引

Production RTL

engrenring/rtl/source/

- board/artty_a7_100t_aes_gcm_uart_rsp_top.v
- aes_gcm_axi_top.v
- stream_core/aes_gcm_stream_core.v
- stream_aes/*.v
- stream_ghash/ghash16.v

Verification

verification/

- sva/ · 37 properties
- uvm/ · UVM 1.2
- results/nist/
- results/board/
- results/exact_current_manifest.json

Vivado native evidence

docs/report/figures/vivado_native/

- 04_routed_timing_summary.rpt
- 09_utilization_hierarchical.rpt
- 14_drc.rpt
- 16_route_status.rpt
- vivado_native_manifest.txt

CURRENT REPOSITORY HEAD

a0e8531e4e886d103b2a51d6b9fae682d50bf451